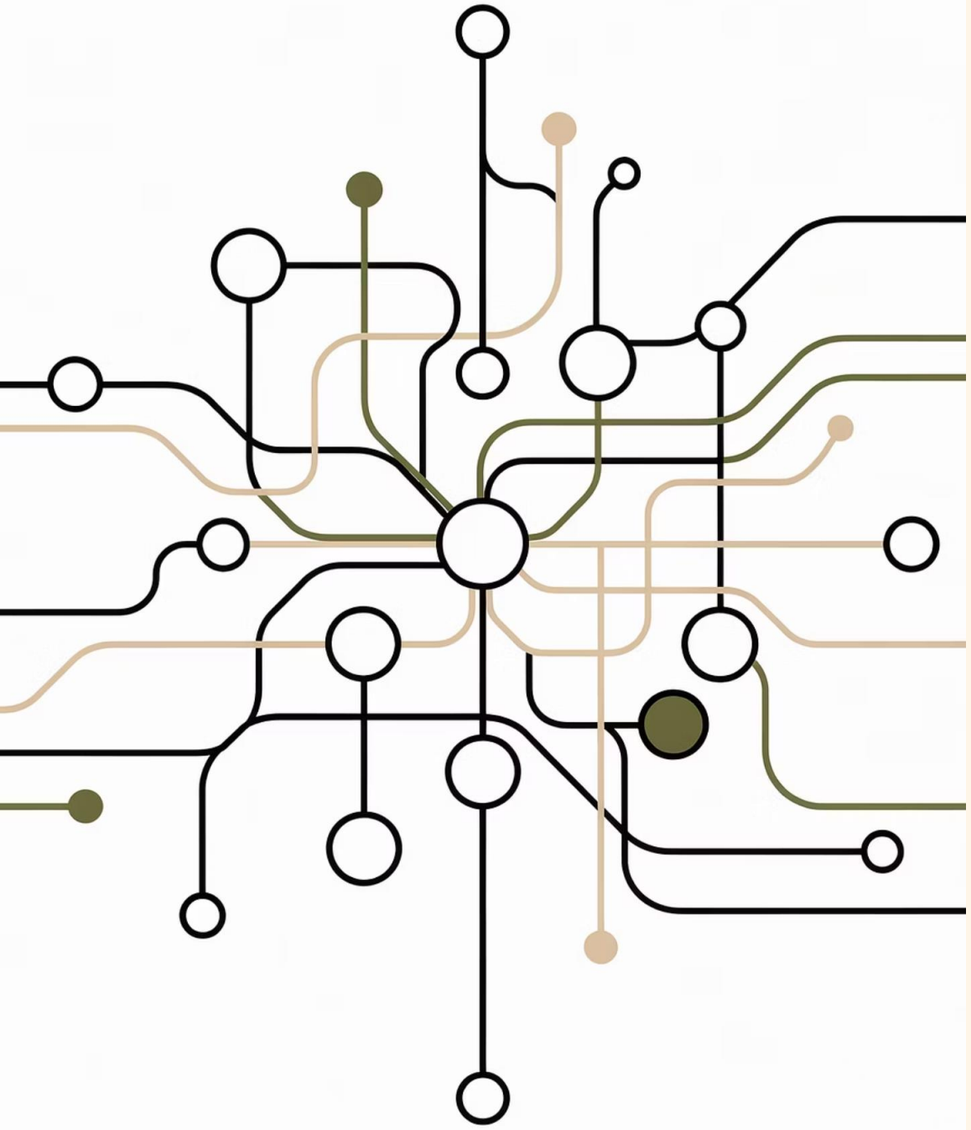


SadakChainAI: Solution Architecture & Roadmap

From Functional MVP to AI-Verified Pilot Deployment (Borivali West)

Creating the "Ground Truth Protocol" for Public Infrastructure.

Current MVP Status (The Foundation)

State: Fully Functional PWA (React + Supabase)

Frontend Interface

Tech Stack: *React (TypeScript) + Vite + Tailwind CSS.*

Status: *Complete. Features responsive mobile UI, location capture, and media handling.*

Backend & Persistence

Tech Stack: *Supabase (PostgreSQL).*

Status: *Complete. Handles User Auth (eKYC type), Role Management (Citizen/Auditor), and secure storage of Raw Reports (photos/videos).*

Logic Layer

Tech Stack: *Edge Functions.*

Status: *Functional. Manages report submission triggers and questionnaire logic.*

- ***Objective:*** *This layer currently serves as the Data Ingestion Point for the upcoming AI/Blockchain integration.*

The Pre-Requisite: Geospatial Node Mapping

Before reporting begins, the physical world is digitized into addressable assets.

1. The Hierarchy:

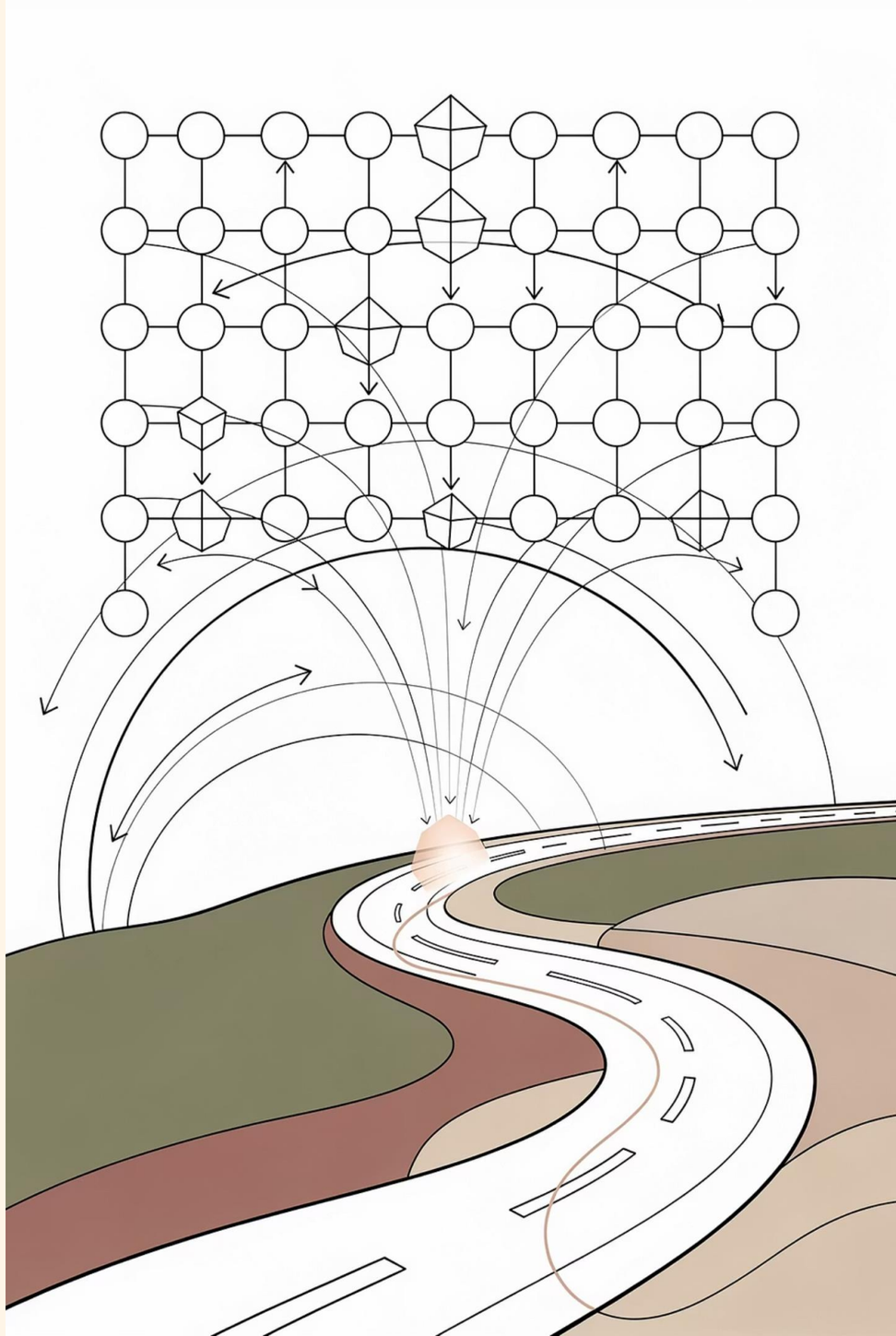
- ***Level 1:*** *Pincode (e.g., Borivali West - 400092).*
- ***Level 2:*** *Nagar/Locality Cluster (e.g., IC Colony).*
- ***Level 3:*** *Road Segment/Chowk (The Atomic Unit).*

2. The "Blockchain Node" Definition:

- *Every **1 KM Road Segment** (or distinct Lane/Chowk) is assigned a **Unique Asset ID (UAID)**.*
- ***Process:*** *We map the Pilot Ward using OpenStreetMap/GIS data, segmenting continuous roads into 1km "blocks."*

3. The Anchor Point:

- *Citizens do not just "report a pothole"; they submit data to a specific **UAID**.*
- *This UAID is the permanent "Container" on the blockchain that holds the history of that specific kilometer.*



Pilot Architecture Workflow (The Data Pipeline)

How data moves from Citizen to Blockchain. (Future Deployment)

01

Ingestion (Current MVP)

*PWA captures photo + GPS. System auto-matches GPS to the nearest **UAID (Road Segment)**.*

02

Perception Layer (New AI)

Data is routed to Data Quality Filter (DQF) and Pothole Damage Classifier (PDC).

03

Decision Layer (Proprietary)

*System synthesizes AI scores to calculate the **Road Trust Score (RTS)** for that UAID.*

04

Trust Layer (New Blockchain)

- *The RTS + Metadata is hashed.*
- *The hash is anchored to the **Permissioned Ledger** under the specific **UAID**.*

05

Execution Layer

Smart Contract linked to that UAID reads the new anchor and triggers payment/fine logic.

AI Technologies Used (The Perception Layer)

Requirement: Real-time verification of noisy data.

1

Data Quality Filter (DQF)

Function: *NLP/ML Classifier to detect spam, gibberish text, and metadata inconsistencies.*

Model Architecture: *Random Forest or Lightweight Transformer (e.g., DistilBERT) for text analysis.*

Role: *The "Gatekeeper" that rejects invalid reports before they consume GPU resources.*

2

Data Quality Filter (DQF)

Function: *NLP/ML Classifier to detect spam, gibberish text, and metadata inconsistencies.*

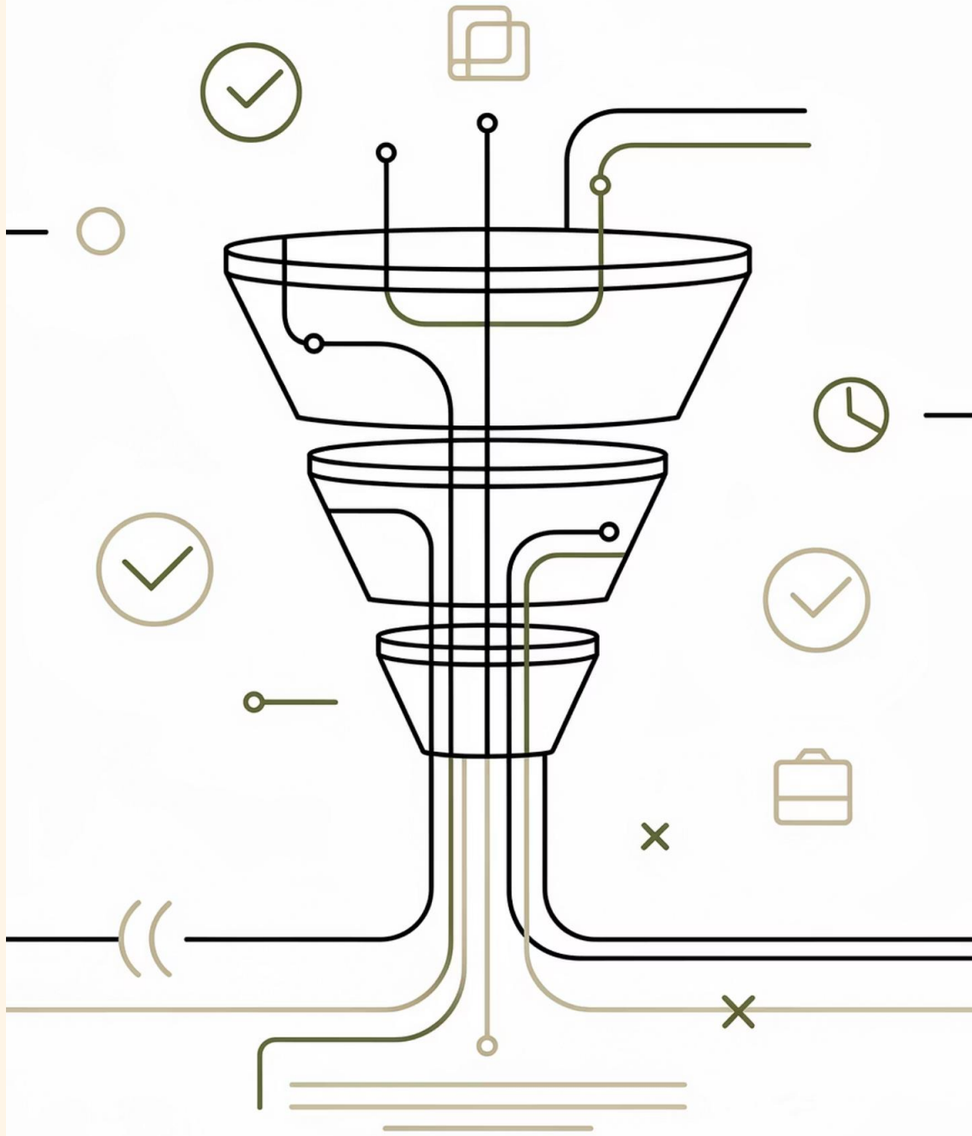
Model Architecture: *YOLOv8 (for speed/detection) or ResNet-50 (for classification depth).*

Output: *Classifies damage type (Pothole, Crack, Waterlogging) and Severity (Low, Medium, Severe).*

3

Large Vision Model (LVM) Foundation

Long-Term Asset: *The pilot begins aggregating data to fine-tune a foundational LVM (e.g., based on LLaVA or similar multimodal models) for future autonomous driving contexts.*



Training Datasets & Sources

The transition from "Noisy Input" to "Immutable Truth".

Dataset A: Raw Citizen Ingest

Source: *MVP PWA*

Content: *Unfiltered photos/videos from Borivali West pilot users.*

Use: *Initial training and stress-testing the DQF model.*

Dataset B: The Immutable Training Pool

Source: *Blockchain*

Content: *The "Gold Standard" reports (anchored to UAIDs) that have passed AI + Human Audit.*

Use: *Retraining the PDC model to prevent drift and ensure high fidelity.*

Dataset C: Base GIS Data

Source: *OpenStreetMap/Google*

Content: *Satellite & Street Maps used to define the initial UAID grid (Slide 3).*

Use: *Contextual validation of road segments.*

Resources Utilized - Open Source

Leveraging community standards for cost-efficiency.

Frontend/App

React, TypeScript, Vite, Tailwind CSS, Lucide React (Icons).

Backend/Database

Supabase (built on open-source PostgreSQL), PostGIS (for geospatial queries).

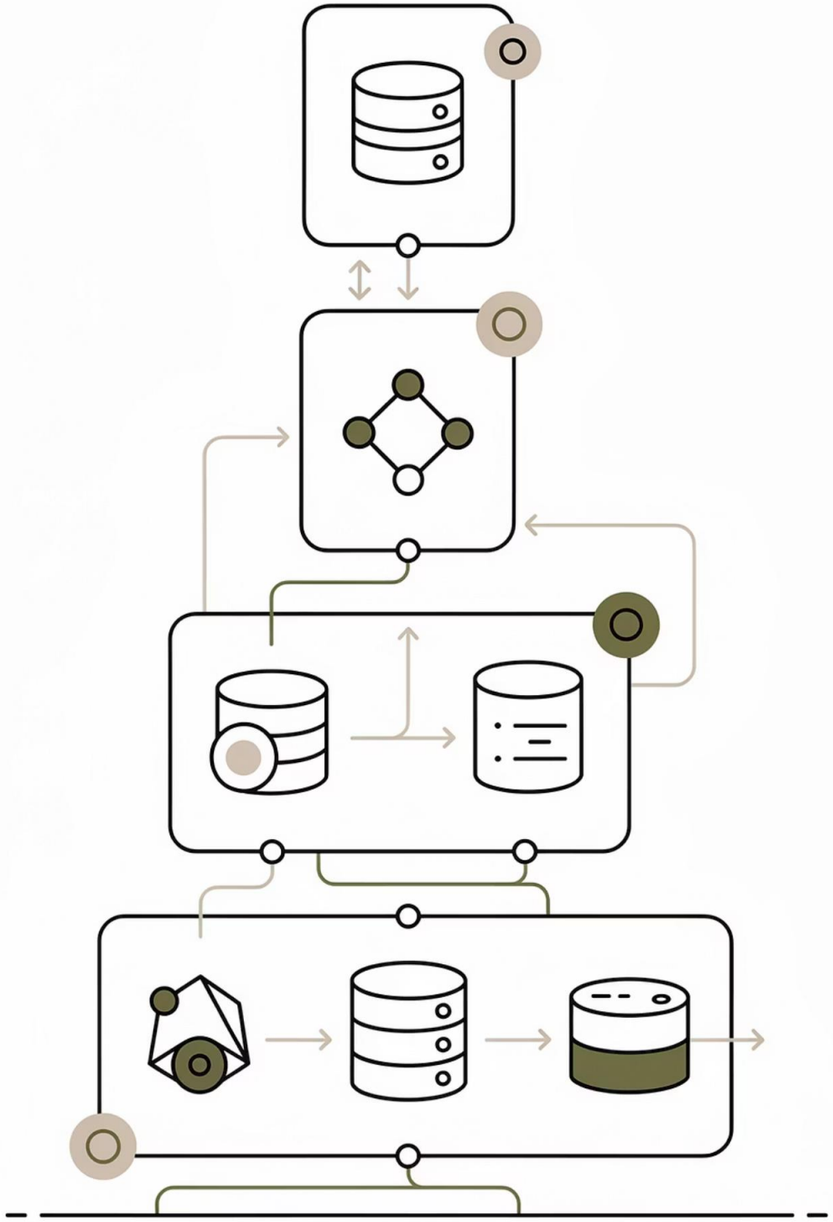
AI/ML Frameworks

TensorFlow / PyTorch: *For model development.*

OpenCV: *For pre-processing image data.*

Blockchain Protocol

Polygon CDK or Hyperledger Besu: For setting up the private/permissioned Proof-of-Authority network.



Resources Utilized - Proprietary (IP)

The core competitive assets of SadakChainAI.

The Geospatial Node Grid

The proprietary heuristic logic that synthesizes DQF scores, PDC visual confidence, and user reputation into a single, actionable 0-100 score.

Trained Model Weights

The specific fine-tuning of the YOLO/ResNet models on Indian road conditions (the "Chaos" Moat).

The Immutable Training Pool

The aggregated, cleaned, and verified dataset itself, which becomes a sovereign national asset.

The Road Trust Score (RTS) Algorithm

The proprietary heuristic logic that synthesizes DQF scores, PDC visual confidence, and user reputation into a single, actionable 0-100 score.

Resources Utilized - Licensed/Commercial

Infrastructure required for the Pilot deployment.

Cloud Compute (OpEx)

AWS/Azure/GCP: *Renting GPU instances (e.g., NVIDIA T4 or A10) for AI inference.*

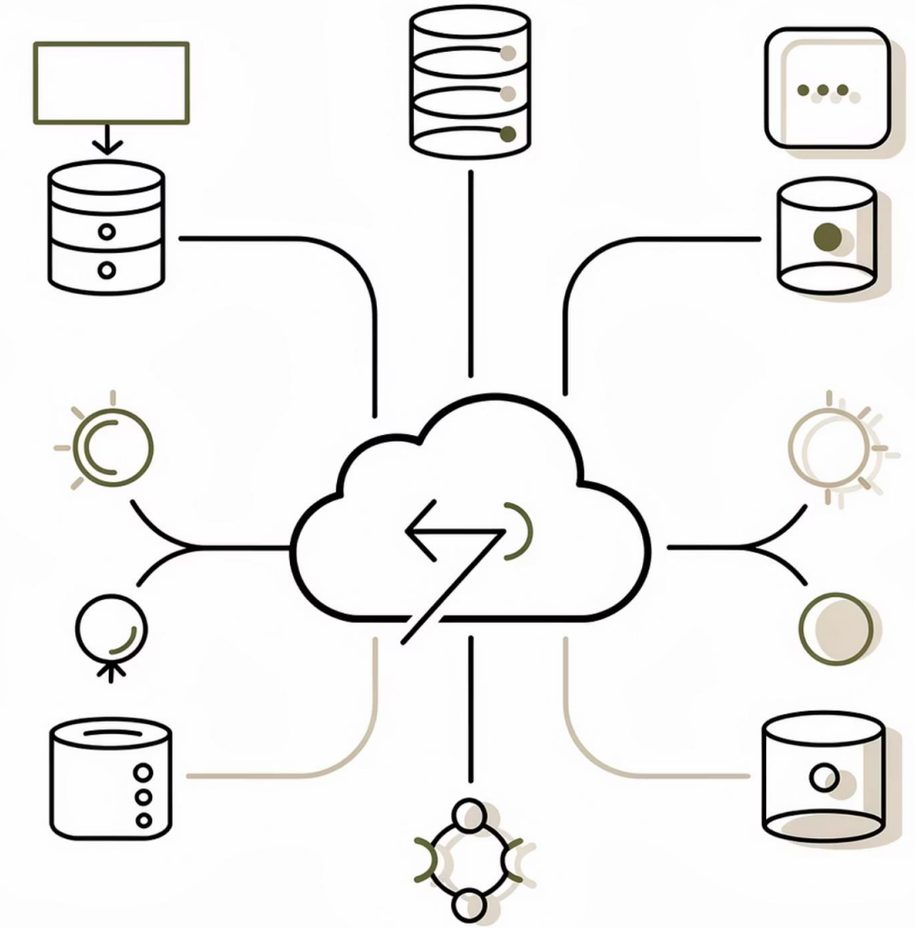
Validator Nodes: *Hosting the 5-node Proof-of-Authority blockchain network.*

Mapping APIs

Google Maps Platform / Mapbox: *Licensed API access for precise geocoding and reverse-geocoding during the reporting process.*

Notification Services

SendGrid: *For transactional email/OTP delivery (Email Confirmation).*



Pilot Deployment Roadmap (Borivali)

Summary of Immediate Next Steps.

Phase 1 (Weeks 1-4)

1

Deploy Blockchain Nodes (AWS). Smart Contract deployment for "Staking" simulation.

Phase 2 (Weeks 5-8)

2

Connect MVP Frontend to AI Inference API (Python/Flask wrapper around models).

Phase 3 (Weeks 9-12)

3

"Shadow Mode" Launch. Citizens report via PWA; AI scores in background; successful reports are anchored to the testnet ledger.

Goal: *Validate the Time-to-Resolution metric and establish the first 1,000 verified data points for the LVM.*